

Database Schema Documentation Template

1. Introduction

1.1. Purpose

[Describe the purpose of this database schema documentation.]

1.2. Scope

[Define the scope of the database system.]

1.3. Database Overview

[Provide a high-level overview of the database architecture.]

2. Database Environment

Component	Technology / Details
Database Engine	[PostgreSQL / MySQL / MongoDB / etc.]
Hosting Environment	[Cloud / On-Premise]
ORM / Query Tool	[SQLAlchemy / Prisma / etc.]
Backup Strategy	[Daily / Weekly / Automated]

3. Schema Overview

Schema Name	Description
public	[Main application schema]

4. Table Definitions

Table Name	Description	Primary Key
users	[Stores user information]	user_id
orders	[Stores order details]	order_id

5. Column Specifications

Column Name	Data Type	Nullable	Default Value	Description
user_id	UUID	No	Generated	[Unique user identifier]
email	VARCHAR(255)	No	NULL	[User email address]

6. Relationships and Constraints

Constraint Type	Table	Description
Primary Key	users	[Primary key for users table]
Foreign Key	orders	[Links orders to users]
Unique Constraint	users	[Ensures unique email addresses]

7. Indexing Strategy

Index Name	Table	Columns	Purpose
idx_users_email	users	email	[Improve email lookup performance]

8. Stored Procedures and Triggers

Name	Type	Description
update_timestamp	Trigger	[Automatically updates modified date]

9. Data Dictionary

Field Name	Business Meaning
status	[Represents current record state]
created_at	[Record creation timestamp]

10.Security and Access Control

Security Area	Implementation
Authentication	[Database authentication mechanism]
Authorization	[Role-based access control]
Encryption	[Encryption at rest / in transit]

11.Backup and Recovery

Backup Type	Frequency	Retention
Full Backup	Daily	[30 Days]
Incremental Backup	Hourly	[7 Days]

12.Performance Considerations

12.1. Optimization Strategy

[Describe query optimization and scaling strategies.]

12.2. Monitoring Tools

[Describe monitoring and performance tracking tools.]

13. Conclusion

[Provide final notes and recommendations.]